

Data fundamentals & unleashing Business Potential using Tableau for Superstore India



TATA STEEL UTILITIES AND INFRASTRUCTURE SERVICES LIMITED

Presented By,

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Duration: 24th May 2025 – 4th July 2025

UNDER THE GUIDANCE OF

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Ms. SHRUTI SHUBHANGEE

ACKNOWLEDGMENT

I would like to express my sincere gratitude to TATA UISL (Utilities and Infrastructure Services Limited) for giving me the opportunity to undertake this internship. It has been a valuable learning experience that helped me strengthen my understanding of data analytics and business intelligence using Tableau.

I am deeply thankful to my mentor Mr. Soumya Chaudhury for his continuous support, guidance, and insightful feedback throughout the internship. I would also like to thank Ms. Shruti Subhangee for her encouragement and assistance during various stages of the project.

Finally, I am grateful to my college, Birla Institute of Technology, Mesra, and all those who directly or indirectly supported me in completing this internship successfully.

TATA STEEL UTILITIES AND INFRASTRUCTURE SERVICES LIMITED
PROJECT COMPLETION REPORT

To be completed by the Trainee:

Date: 04/07/2025

ID Card No: 87

Name of Trainee: RAVI RAUSHAN SINGH

Gender: Male/Female

Institute Name: BIT MESRA

Project Duration: From 22.5.25 to 04.07.25

Course/Discipline: CSE

Entry Recommended by: Self/ TSUISL employee... T.S.U.I.S.L....

Project title: Data Fundamentals & unleashing Business Potential using Tableau

Following points will be completed by the project guide/ HOD/ recommender of Intern:

1. Is Project Work linked to departmental KPI/ Jobs improvement plan? NA
2. Trainees add value through project work for sectional KPI/ customers satisfaction? NA
3. Project Works have added value in terms of saving time cycle / Improve safety/ ease to work/ save money/ resource (Manpower/other)/ Improve KPI/ any others specific things NA
4. If this work is outsourced how much cost involved (approx.) NA

GUIDE DETAILS

NAME: Shruti Shubhangee

DEPT: Technical Services

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Feedback about the impact for his section/ dept. completed by Intern/ Trainee:

NA

Suggestion for Improvements by the Guide.

As discussed with Mr. Ravi.

Suggestion for Improvement by the Trainee.

The above project has been successfully completed. He/ She has cleared all the dues related to project work.

Ravi Raushan Singh

Signature of the Trainees

Shruti Shubhangee

Signature of Project Guide (with seal)

04/07/25

Suparna Dutta

Signature of Manager

INTRODUCTION

This report summarizes my 6-week internship at TATA UISL (Utilities and Infrastructure Services Limited), which began on 24th May 2025, under the guidance of Mr. Soumya Chaudhury and Ms. Shruti Subhangee. The focus was on "Data Fundamentals & Unleashing Business Potential Using Tableau for Superstore India."

During the internship, I worked with the Superstore India dataset to understand data analytics concepts and create interactive dashboards in Tableau. The goal was to transform raw data into meaningful insights, highlighting sales trends, regional performance, and customer behavior to support better business decisions.

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1.Statistics

i. Fundamentals of Statistics: -

a) Population: -

All of the individuals in a group on which you try to draw a conclusion.

Example - All 5000 students in a university. We want to know the average height of all of them.

b) Sample: -

The group of persons chosen to represent the population in a statistical study.

Example - We measure the height of 100 randomly chosen students. This smaller group is our sample.

c) Parameter: -

A quantity which measures something about a population.

Example - The actual average height of all 5000 students. It is a fixed value, but usually unknown.

d) Statistic: -

A quantifier that measures some sample property.

Example - The average height of the 100 students we measured. We use it to estimate the population parameter.

e) Variable:-

A characteristic of an item or an individual that will be analyzed using statistics.

Example - It tells us how spread out the heights are. High variance means big differences, low variance means values are close.

The Branches of Statistics

Two branches, descriptive statistics and inferential statistics, comprise the field of statistics.

a) Descriptive Statistics: -

The branch of statistics that focuses on collecting, summarizing, and presenting a set of data.

Example - The average score of 100 students in the test was 75.

b) Inferential Statistics: -

The branch of statistics that analyzes sample data to draw conclusions about a population.

Example - Based on the sample, we estimate that the average score of all students in the school is around 75.

Sampling Concepts

a) Sampling

The process by which members of a population are selected for a sample.

Example - We select 100 customers from a list of 1,000 to study their buying behavior. This small group is the sample used to represent the whole population.

b) Probability Sampling

A sampling process that takes into consideration the chance of occurrence of each item being selected. Probability sampling increases your chances that the sample will be representative of the population.

Example - Each customer has a known and equal chance of being selected. This makes the sampling process fair and unbiased.

c) Simple Random Sampling

The probability sampling process in which every individual or item from a population has the same chance of selection as every other individual or item. Every possible sample of a certain size has the same chance of being selected as every other sample that has that size.

Example - We randomly pick 50 students from a class of 200 using a lottery method. Every student had an equal chance of being chosen.

d) Frame

The list of all items in the population from which samples will be selected.

Example - A list of all 500 registered voters in a city.

ii. Presenting Data in Charts and Tables:

Presenting Categorical Data

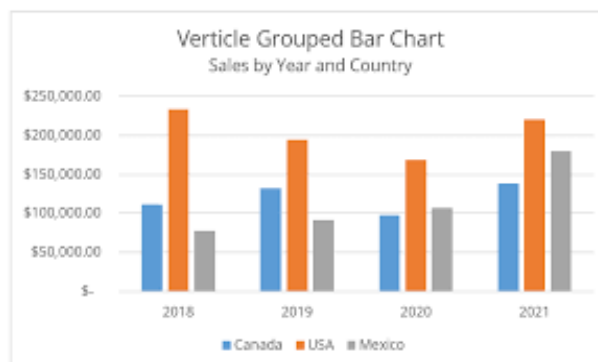
a) The Summary Table

A two-column table in which the names of the categories are listed in the first column and the count, amount, or percentage of responses are listed in a second column. Sometimes additional columns present the same data in two or more ways (for example, as counts and percentages).

Behaviour	Percentage (%)
Donate regularly	11
Have donated and may again	31
Have not but may in future	17
Never have, never will	13
Want to, but think they cannot	28

b) Bar Graph

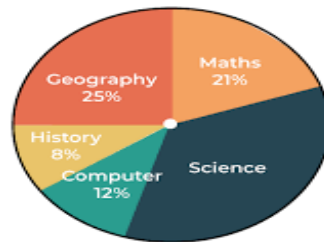
A Bar graph is a type of data-handling method that is popularly used in statistics. A bar graph or bar chart is a visual presentation of a group of data that is made up of vertical or horizontal rectangular bars with lengths that are equal to the measure of the data.



c) Pie Chart

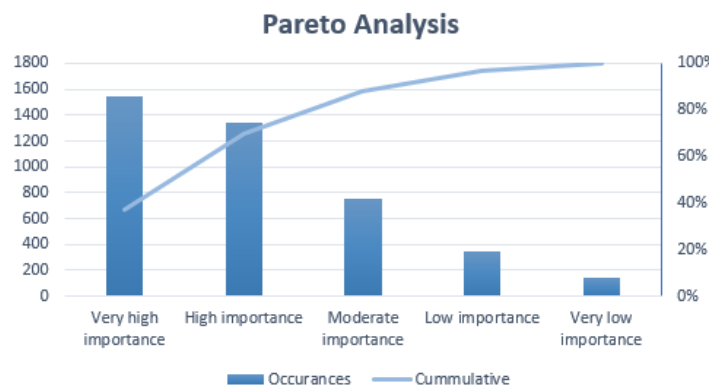
Pie chart is a popular and visually intuitive tool used in data representation, making complex information easier to understand at a glance. This circular graph divides data into slices, each representing a proportion of the whole, allowing for a clear comparison of different categories making it easier.

Number of Students



d) The Pareto Diagram

A Pareto chart is a bar graph. The lengths of the bars represent frequency or cost (time or money), and are arranged with longest bars on the left and the shortest to the right. In this way the chart visually depicts which situations are more significant.



e) Two-Way Cross-Classification Tables

A multicolumn table that presents the count or percentage of responses to two categorical variables. In two-way tables, the categories of one of the variables form the rows of the table, while the categories of the second variable form the columns. Cross-classification tables are also known as cross-tabulation tables.

Two-Way Table



Statistical table that shows the observed number or frequency for two variables.

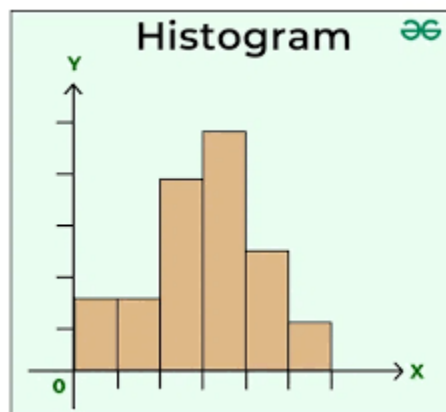
	Maths	English	Total
Internal	3	4	7
External	7	9	16
Total	10	13	23

Presenting Numerical Data

a) Histogram

A histogram is a graph that shows the distribution of numerical data using bars. Each bar represents the frequency of data within a certain range (called bins). It helps us understand the shape and spread of data.

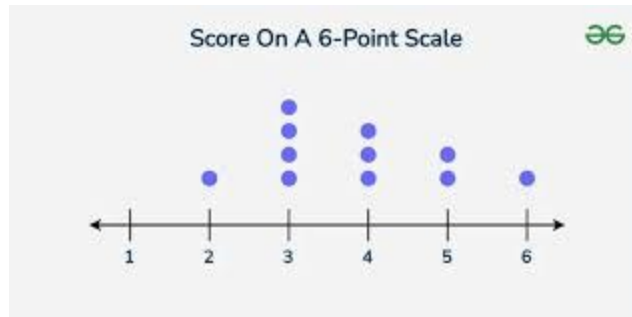
Example: A histogram of student marks shows how many students scored within 0–10, 10–20, etc.



b) The Dot Scale Diagram

A dot plot displays individual data points using dots along a number line. Each dot represents one occurrence of a value. It's useful for small datasets to see frequency clearly.

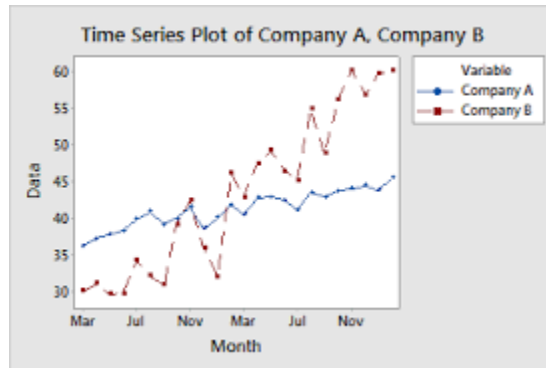
Example: A dot plot showing how many students scored 5, 6, 7 marks in a quiz.



c) The Time-Series Plot

A time-series plot shows how data changes over time. Time is on the x-axis and values on the y-axis. It's great for spotting trends and patterns.

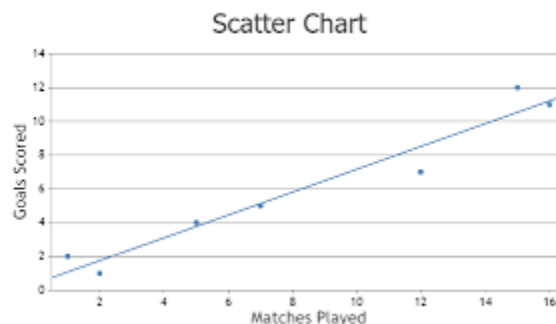
Example: A line graph of monthly sales over a year.



d) The Scatter Plot

A scatter plot shows the relationship between two numerical variables. Each point represents one data pair (x, y). It helps identify correlations or patterns.

Example: A scatter plot of hours studied vs. exam scores.



iii. Descriptive Statistics for Numerical Variables

Measures of Central Tendency

a) The Mean

The mean is the average of a set of numbers. Add all the values and divide by how many values there are.

Example: For 10, 20, 30 $\rightarrow$ Mean = $(10 + 20 + 30)/3 = 60/3 = 20$

b) The Median

The median is the middle value in a sorted list. If the count is odd, it's the centre number; if even, average the two middle numbers.

Example: For 10, 20, 30 $\rightarrow$ Median = 20

c) The Mode

The mode is the number that appears most often.

Example: For 10, 20, 20, 30 $\rightarrow$ Mode = 20

d) Quartiles

It divide a sorted dataset into four equal parts.

Q1 (First Quartile): 25% of data lies below this value. Example: In [10, 20, 30, 40, 50], Q1 = 20

Q2 (Second Quartile / Median): 50% of data lies below this. Example: In [10, 20, 30, 40, 50], Q2 = 30

Q3 (Third Quartile): 75% of data lies below this value. Example: In [10, 20, 30, 40, 50], Q3 = 40

Measures of Variation

a) The Range

Range is the difference between the highest and lowest values in a dataset.

Example: In [10, 15, 20], range = $20 - 10 = 10$.

b) Standard (z) Score

Standard Score (Z-score) tells how far a value is from the mean, in standard deviations.

Formula: $Z = (X - \text{Mean}) / \text{Standard Deviation}$.

Example: If a test score is 85, mean = 75, SD = 5 $\rightarrow Z = (85 - 75)/5 = 2$.

Shape of Distributions

a) Right Skewed Shape (Positively Skewed)

Most data is on the lower side, with a long tail on the right. Mode < Median < Mean.

Skewed towards higher values.

Example: Income distribution (few people earn very high).

b) Symmetrical Shape

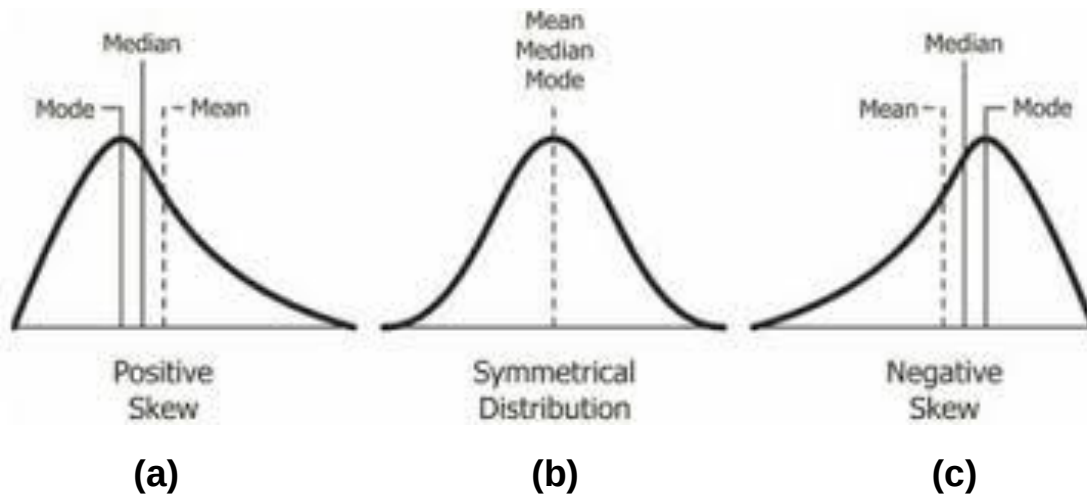
A symmetrical shape has data evenly spread on both sides of the center. Mean, median, and mode are nearly equal. Looks like a bell curve (normal distribution).

Example: Heights of students in a class.

c) Left Skewed Shape (Negatively Skewed)

Most data is on the higher side, with a long tail on the left. $\text{Mean} < \text{Median} < \text{Mode}$. Skewed towards lower values.

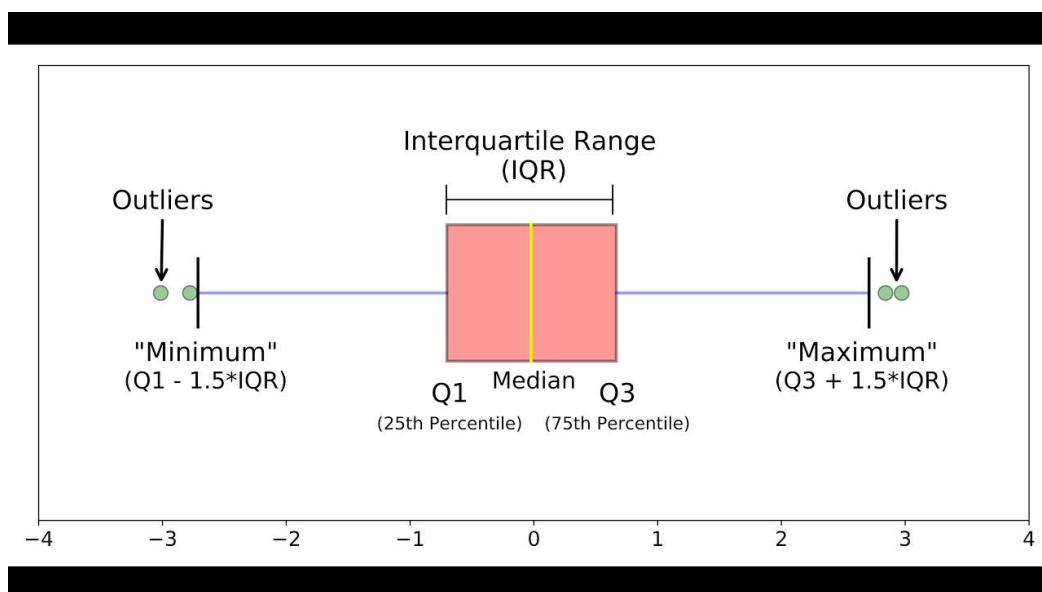
Example: Age at retirement (few retire early).



d) Box and Whisker Plot

A chart that shows the spread and center of data using 5 numbers. Displays minimum, Q1, median, Q3, and maximum. Helps identify outliers and skewness.

Example: Test scores comparison between two classes.



iv. Probability

a) Random Variable

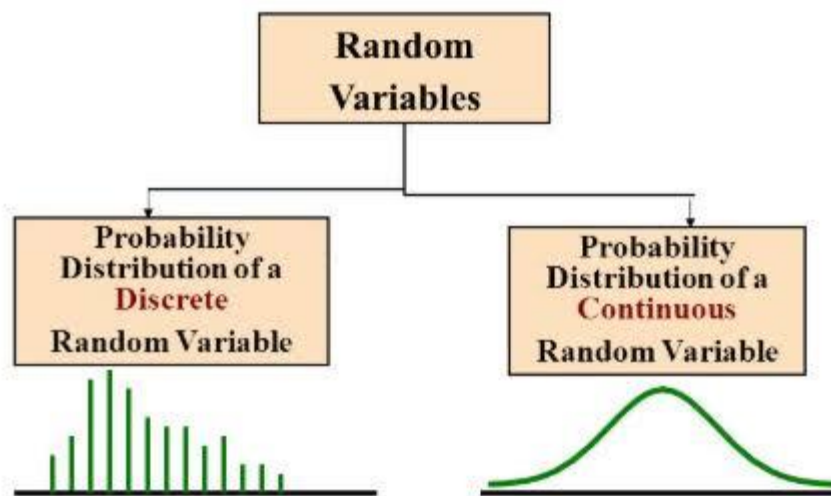
A random variable is a variable that takes different values based on the outcome of a random event. It can be discrete (countable) or continuous (measurable). It assigns a number to each outcome in a sample space.

Example: In a coin toss, $X = 1$ for heads, $X = 0$ for tails.

b) Probability Continuous Probability Distributions

Probability measures how likely an event is to happen. It ranges from 0 (impossible) to 1 (certain). The formula is: $\text{Probability} = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$

Example: Probability of getting heads in a coin toss = $1/2$.



v. Probability Distributions

a) Discrete Probability Distribution

A discrete probability distribution lists the probabilities of all possible outcomes of a discrete random variable. Example: Tossing a coin — $P(\text{Heads}) = 0.5$, $P(\text{Tails}) = 0.5$.

Each outcome has a probability between 0 and 1, and they all add up to 1.

b) The Expected Value of a Random Variable

The expected value is the average outcome if a random experiment is repeated many times. Example: Rolling a fair die — $E(X) = (1+2+3+4+5+6)/6 = 3.5$.

It tells what value you "expect" on average over many trials.

c) Standard Deviation of a Random Variable (σ)

Standard deviation measures how spread out the values are from the expected value. Example: If $E(X) = 3.5$ in a die roll, σ tells how much typical rolls vary from 3.5.

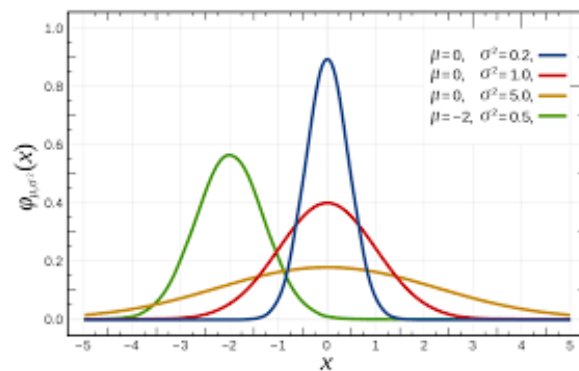
Lower σ means outcomes are close to the average; higher σ means more spread.

vi. Continuous Probability Distributions and the Normal Distribution

a) Continuous Probability Distributions

It deals with variables that can take any value in a range, like height or time. The probability is represented by the area under a curve (not exact values).

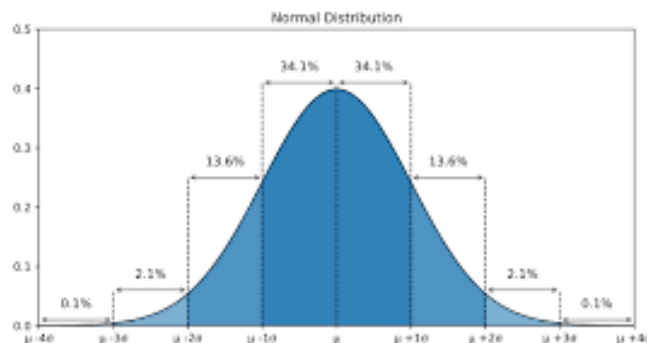
Example: Time taken by students to finish a test follows a continuous distribution.



b) Normal Distribution

It is a bell-shaped symmetric curve where most values lie around the mean. Mean = Median = Mode, and data is evenly spread.

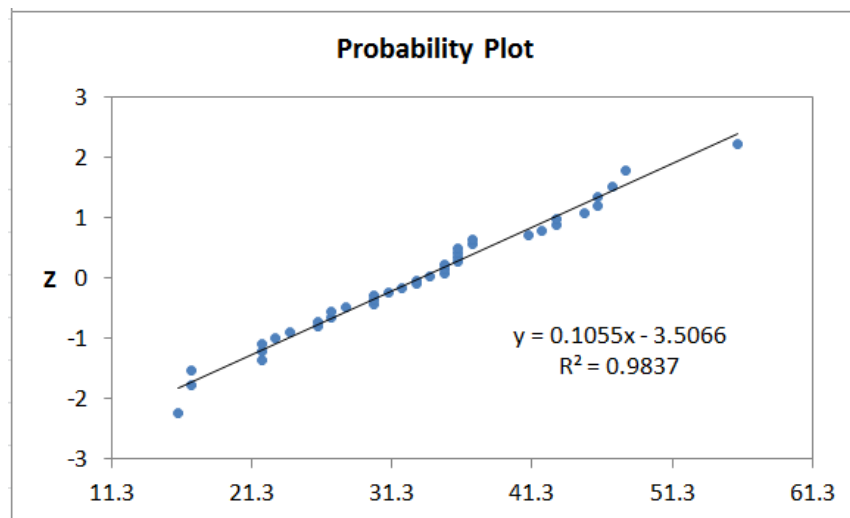
Example: Heights of people in a population often follow a normal distribution.



The Normal Probability Plot

A Normal Probability Plot compares your data with a normal distribution. If the points form a straight line, the data is likely normal. Deviations from the line show skewness or outliers.

Example: You plot the heights of 100 students. If the points fall roughly along a straight line, the heights are normally distributed.

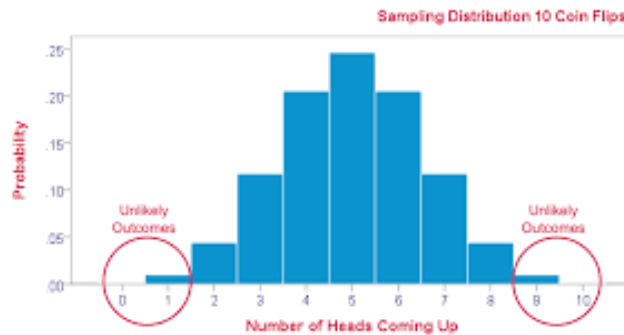


Sampling Distributions and Confidence Intervals

a) Sampling Distributions

It is the distribution of a statistic (like mean) from many samples of the same size.

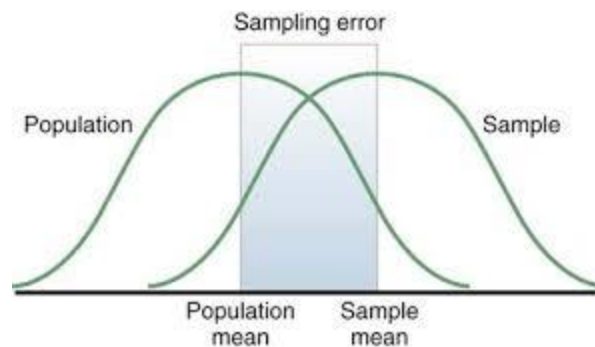
Example: Taking average marks from 100 students in different random groups and plotting their means.



b) Sampling Error

It is the difference between a sample statistic and the true population value.

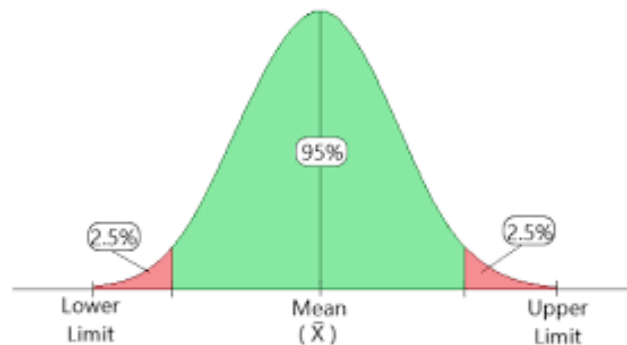
Example: If the actual average height is 160 cm, and your sample average is 158 cm, the error is 2 cm.



c) Confidence Interval Estimate for the Mean (using t-distribution)

It gives a range to estimate the population mean when the sample size is small and population standard deviation is unknown.

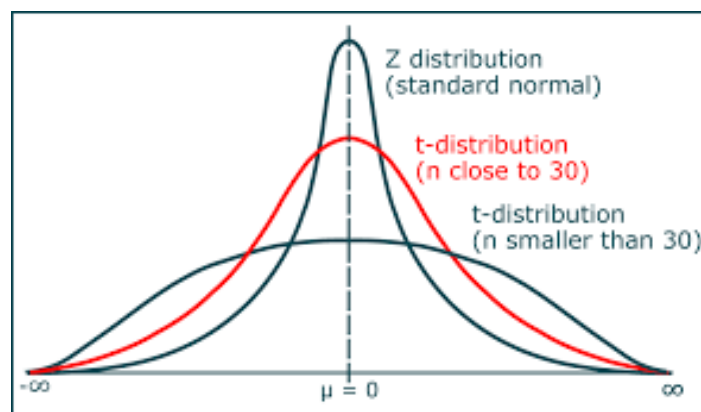
Example: A sample of 20 students has an average score of 75 with a standard deviation of 10; the 95% CI gives a range like (71.6, 78.4).



d) t-Distribution

It is a probability distribution used to estimate population parameters when the sample size is small and population standard deviation is unknown.

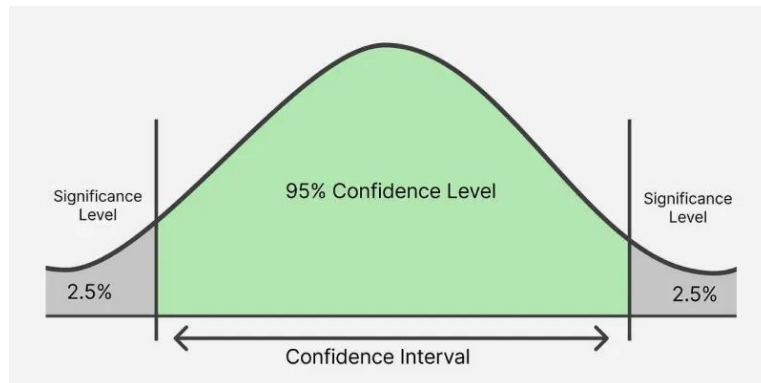
Example: If you have a sample of 15 students' test scores and want to estimate the average score of all students, use t-distribution.



e) Confidence Interval Estimation for the Proportion

It gives a range to estimate the true proportion of a population based on a sample.

Example: If 60 out of 100 people prefer tea, the 95% CI for tea preference proportion might be (0.51, 0.69).



vii. Hypothesis Testing

a) Null Hypothesis (H_0)

It is a statement that there is no effect or no difference. We assume it is true until proven otherwise.

Example: H_0 : A new medicine has no effect on patients.

b) Alternative Hypothesis (H_1)

It is a statement that contradicts the null hypothesis. It suggests there is an effect or difference.

Example: H_1 : The new medicine improves patient health.

c) Hypothesis Testing

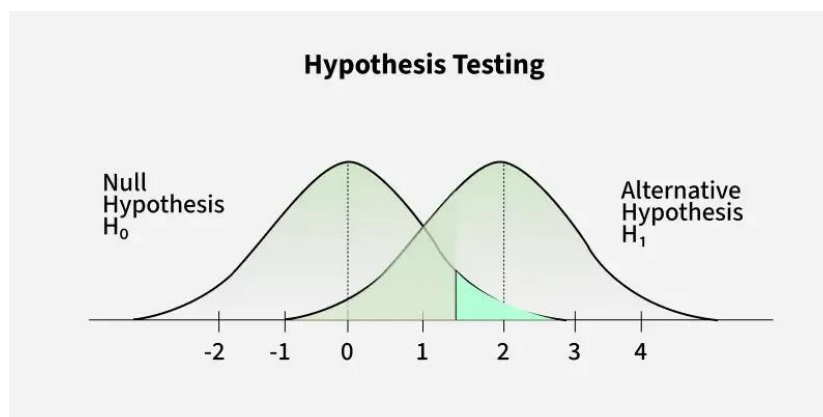
It is the process of using data to accept or reject the null hypothesis. We use sample data to make decisions about a population.

Example: Testing if exam scores improve after a new teaching method.

d) Test Statistic

It is a number calculated from the data to test the hypothesis. It helps decide whether to reject H_0 .

Example: A t-score or z-score used in a statistical test.



Decision-Making Risks Testing for the Difference Between Two Proportions

a) Type I Error (False Positive)

Rejecting a true null hypothesis.

Example: A doctor says you have a disease, but you are actually healthy.

b) Type II Error (False Negative)

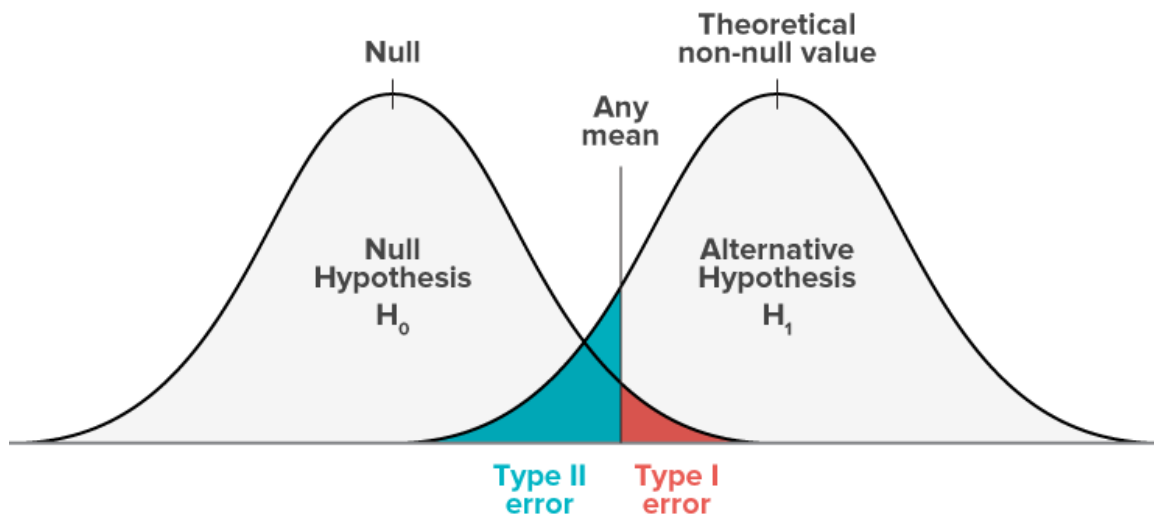
Failing to reject a false null hypothesis.

Example: A test says you're healthy, but you actually have the disease.

c) Risk Trade-Off

Reducing one type of error increases the other.

Example: Making a medical test more sensitive may reduce false negatives but increase false positives.

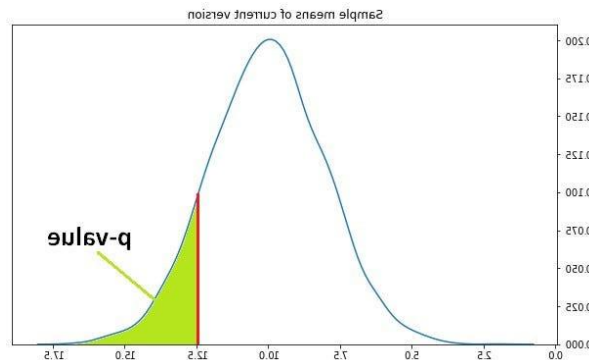


Performing Hypothesis Testing

a) p-Value

The p-value shows the probability that the observed result happened by chance. A smaller p-value (like < 0.05) means the result is statistically significant.

Example: If $p = 0.03$, there's a 3% chance the result is random — we can reject the null hypothesis.



Types of Hypothesis Tests

a) Number of Groups

It refers to how many distinct categories or samples are being compared in a study.

Example: Comparing average sales in 3 different regions means there are 3 groups.

b) Relationship Stated in Alternative Hypothesis (H1)

H1 suggests there is a significant effect or difference between variables or groups.

Example: "There is a difference in sales between North and South regions."

c) Type of Variable

Variables can be categorical (like Region, Gender) or numerical (like Sales, Profit).

Example: "Region" is categorical; "Sales" is numerical.

viii. Hypothesis Testing: Z and t Tests

a) Hypothesis Testing:

It is a statistical method to decide if there is enough evidence to support a specific claim about a population based on sample data.

Example: A company claims its battery lasts 10 hours. A test on 30 batteries shows an average of 9.5 hours. Hypothesis testing checks if this difference is significant or due to chance.

Z-Test :

Used when sample size is large ($n > 30$) and population standard deviation is known.

Example: Testing if average height of students is 170 cm using known population SD and a large sample.

T-Test :

Used when sample size is small ($n \leq 30$) or population standard deviation is unknown.

Example: Comparing average test scores of 15 students to a known average, without knowing population SD.

b) Testing for the Difference Between Two Proportions

This test checks if two population proportions are significantly different. We compare the proportion of success in two different groups. Used when data is categorical.

Example: Comparing the % of voters preferring Candidate A in two cities.

c) Testing for the Difference Between the Means of Two Independent Groups

This test checks if the average values (means) of two separate groups are different. Used when the groups are unrelated and data is numerical.

Example: Comparing average exam scores of students from two different schools.

ix. Hypothesis Testing: Chi-Square Tests and the One-Way Analysis of Variance (ANOVA)

a) Chi-Square Test for Two-Way Tables:

It checks whether two categorical variables are related or independent. Used on a contingency table (cross-tab) showing frequency counts. Compares observed vs. expected counts to find statistical significance.

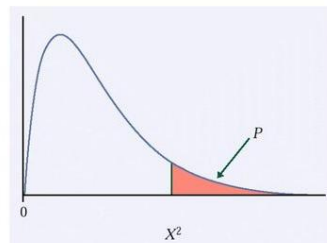
Example: A survey records gender (male/female) and product preference (A/B). Chi-square test checks if product preference is linked to gender. If p-value < 0.05, there's a significant association.

H_0 : there is no association between the row and column variables

H_a : H_0 is not true

The χ^2 statistic is summed **over all $r \times c$ cells** in the table:

$$\chi^2 = \sum \frac{(\text{observed count} - \text{expected count})^2}{\text{expected count}}$$



P-value: $P(\chi^2 \text{ variable} \geq \text{calculated } \chi^2)$

When H_0 is true, the χ^2 statistic follows $\sim \chi^2$ distribution with **$(r-1)(c-1)$ degrees of freedom.**

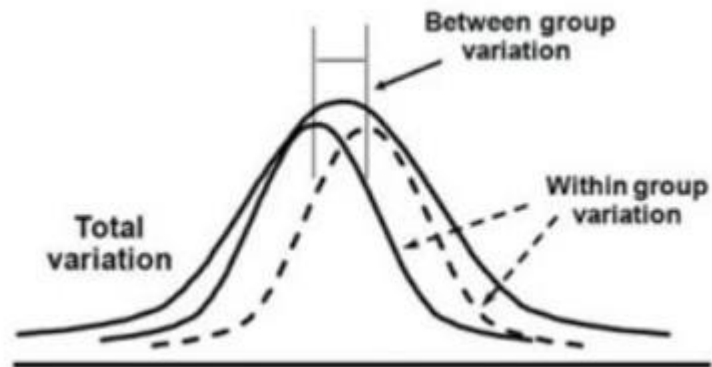
b) One-Way ANOVA:

It compares means of three or more groups based on one factor. Checks if at least one group mean is significantly different. Assumes equal variances and normal distribution.

Example: Comparing average exam scores of students from 3 different colleges. One-Way ANOVA tests if the college affects average performance. If $p\text{-value} < 0.05$, at least one college mean is different.

c) The Three Variances of ANOVA:

- Between-group variance – variation due to the factor (treatment).
- Within-group variance – variation within each group (random error).
- Total variance – the sum of between and within-group variances.



2.Architecture and Data Processing

a) Data Architecture:

Data architecture defines the structure, storage, and flow of data across systems. It includes how data is collected, stored, integrated, and accessed. A good architecture ensures data is consistent, secure, and accessible for analysis and reporting.

b) ETL (Extract, Transform, Load):

ETL is a core process in data management. It involves extracting data from various sources, transforming it into a usable format, and loading it into a data warehouse or database. ETL helps consolidate data for analytics and decision-making.

c) Data Collection:

This is the process of gathering raw data from different sources such as databases, APIs, files, sensors, or user inputs. Accurate and timely collection ensures reliable analytics. Data can be collected in real-time or in batches depending on business needs.

d) Data Extraction:

Once data is collected, extraction involves pulling relevant data from its source. This may include querying databases, accessing cloud storage, or scraping websites. It is the first step of the ETL process and sets the foundation for further processing.

e) Data Structuring:

Raw data is often unorganized and inconsistent. Structuring organizes this data into predefined formats like tables or schemas. This makes the data easier to store, query, and analyze, especially in relational databases or structured files.

f) Data Formatting:

After structuring, data is formatted to ensure consistency and compatibility. This includes standardizing date formats, currency, units, or text casing. Proper formatting is crucial for accurate data integration, visualization, and reporting.

Superstore Data Architecture – Understanding the Flowchart

1. KPIs, Business Goals, and Vision

This is the foundation of the architecture.

- **KPIs (Key Performance Indicators):** Sales, Profit, Return Rate, Order Processing Time, etc.
- **Business Goals:** Improve delivery times, increase customer satisfaction, optimize profit margins.
- **Vision:** Use data to drive decisions that align with company objectives.

2. Data Sourcing

These are the original inputs coming from real-time or batch data producers.

- **Customer Orders:** Comes from the Orders sheet in Superstore_IND.xlsx.
Fields: Order ID, Order Date, Ship Date, Customer ID, Segment, etc.
- **Transaction Details:** Involves financial and logistical details from the Orders and Returns sheets.
Fields: Sales, Quantity, Discount, Profit, Return Status.
- **Customer Info:** Contains demographic and contact info, likely in the Customers section (or part of the Orders data).
Fields: Customer ID, Customer Name, Region, City, Postal Code.

3. Data Integration

This is the transformation stage.

- Combines all data sources into a structured, cleaned, and enriched format.
- Handles ETL (Extract, Transform, Load) processes.

- Converts raw data into a unified dataset (e.g., joins Orders with Customer Info and Returns).
- May add derived fields like Profit Ratio = Profit / Sales or Return Rate.

4. DW or Data Lake (Big Data)

DW (Data Warehouse): Structured storage with cleaned and processed data.

Data Lake: Used for unstructured or semi-structured data (e.g., feedback, logs, return comments).

5. Data Marts (Data Mart 1 & 2)

Subsets of the Data Warehouse for specific business needs.

Data Mart 1 (e.g., Sales & Profit Analysis)

Fields used: Region, State, Product Name, Sales, Profit

Supports dashboards like:

- Regional sales trend
- Most profitable products
- High-sales cities

Data Mart 2 (e.g., Returns)

Fields used: Order ID, Ship Mode, Return Status, Shipping Time

Supports:

- Return rate by category
- Delayed shipping insights

6. Data Consumption

This is where end users interact with data.

Reports :-

Static summaries: PDF, Excel exports

For management and business teams

Examples:

Monthly region-wise sales

Top 10 profitable customers

Analytics :-

Interactive dashboards in Tableau/Power BI

Drill-down filters, graphs, maps

Examples:

- Analyze Profit Ratio by State
- Track monthly Return Rate trends
- Compare forecasted vs actual sales

Data Consumers

Internal Teams: Sales, Marketing, Logistics

Leadership: Strategic planning

Executives: Make KPI-based decisions

3.Data Analysis

a) Data and Types:

Data refers to raw facts and figures collected for analysis. It is mainly categorized into two types:

Categorical Data – Qualitative information like gender, region, or product category.

Numerical Data – Quantitative data like sales amount, profit, or age.

b) Importance and Use:

Data analysis helps in converting raw data into useful insights. It supports informed decision-making, improves business performance, identifies trends, and solves problems. In today's digital world, data-driven strategies give businesses a competitive edge.

c) Business Cases:

Data analysis is widely used across industries.

Retail: Understand customer behavior, segment customers, optimize inventory.

Finance: Detect fraud, analyze risk, improve investment strategies.

Healthcare: Monitor patient health trends, optimize resource usage.

4. Tableau

a) Why Tableau?

Tableau is a powerful data visualization and business intelligence tool that helps users turn data into interactive and shareable dashboards. It is user-friendly, supports drag-and-drop features, and connects easily to various data sources. Tableau enables faster and deeper data insights, making it ideal for both beginners and professionals.

b) How to Use Tableau?

Using Tableau involves a few simple steps:

1. Connect to data sources like Excel, SQL, or cloud platforms.
2. Drag and drop fields to create visualizations like bar charts, maps, or line graphs.
3. Use filters, parameters, and calculated fields to explore data in depth.
4. Create dashboards to combine multiple visualizations for storytelling.
5. The interface is intuitive, making it easy to analyze and present data effectively.

c) Dimensions and Measures:

In Tableau, data fields are categorized as:

Dimensions – Qualitative data used for categorizing, such as Region, Product, or Date.

Measures – Quantitative data used for calculations, like Sales, Profit, or Quantity.

Dependent and Independent Fields in Superstore India

a) Dependent Fields

These are result fields in which their values depend on other factors in the dataset:

Sales	depends on Quantity, Discount, Unit Price
Profit	depends on Sales, Discount, Shipping Cost
Shipping Cost	may vary based on transportation, location
Discount	may be influenced by product, supplier, or order size

b) Independent Fields

These are input or category fields which don't depend on other fields:

1. Ship Date	2. Order Date
3. Priority	4. Customer Name
5. Product ID	6. Product Name
7. Category	8. Sub-Category
9. Product Container	10. Supplier
11. Order ID	12. Unit Price
13. Product Description	14. Transportation
15. Quantity	16. Expected Date

These fields are manually entered or ordered, not derived.

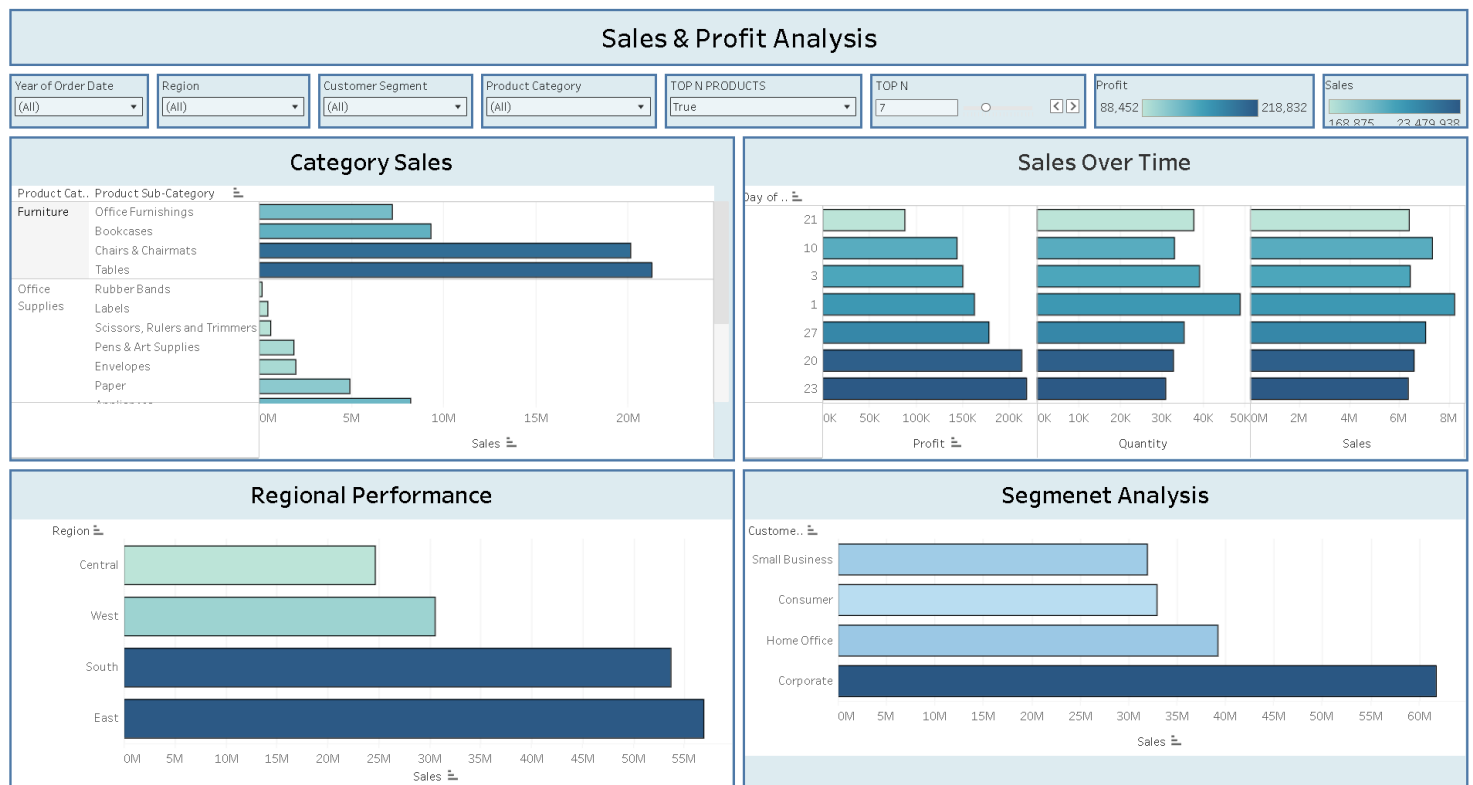
5.Data fundamentals & unleashing Business Potential using Tableau for Superstore India

1. Sales and Profit Analysis

Problem: Understand the sales and profit trends across regions, categories, and customer segments.

Tasks:

- Create dashboards showing total sales, profit, and quantity trends over time.
- Compare sales performance across product categories and sub-categories.
- Highlight top performing and underperforming regions and segments.
- Use filters and parameters for interactive analysis.



INSIGHTS: -

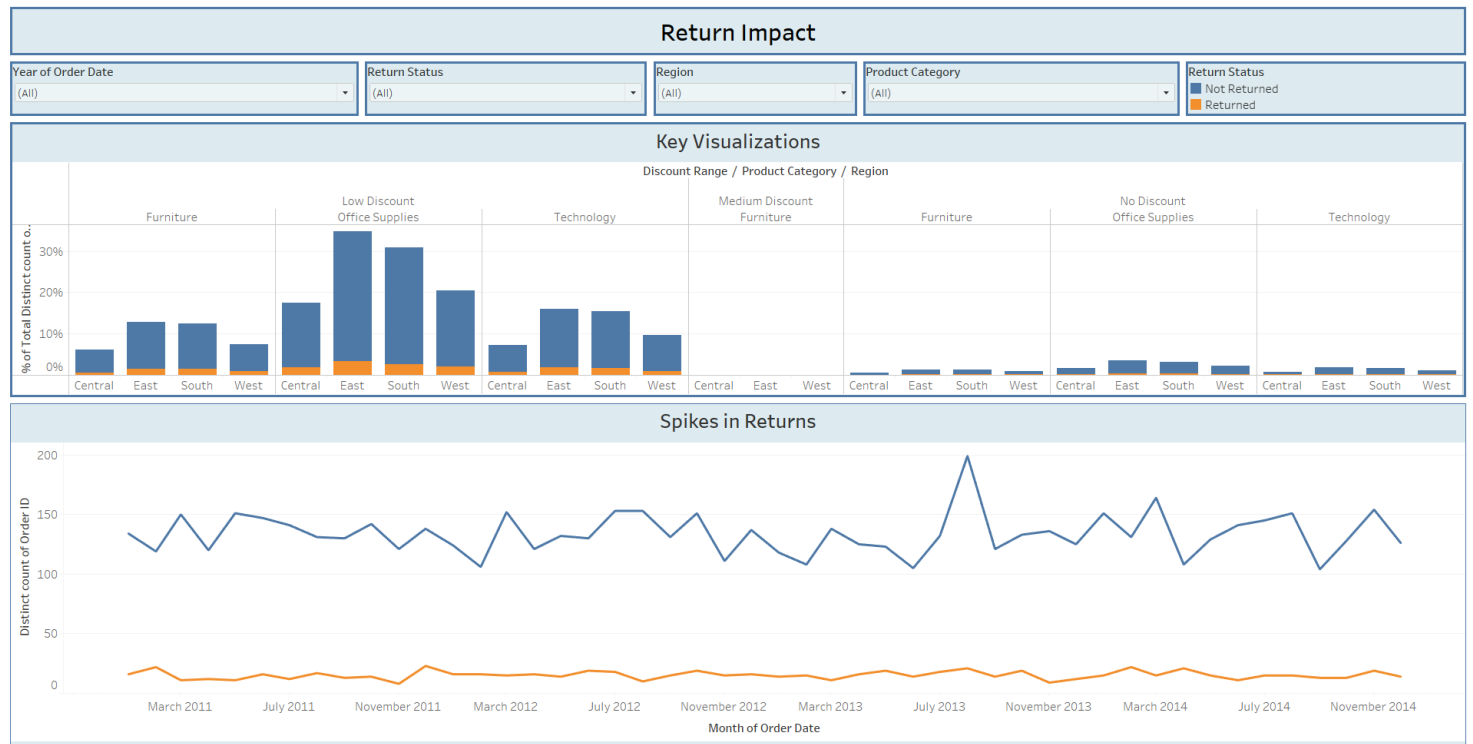
1. The East region clearly stands out, leading in both sales and profit, making it the top - performing area in the country.
2. Corporate segment brings in the highest revenue, crossing ₹75 million, and shows strong profit margins—proving to be our most valuable customer group.
3. When it comes to product categories, Tables and Chairs top the list, generating the most revenue among all sub-categories in Furniture.
4. We see major sales peaks on the 10th and 23rd of each month, suggesting possible customer buying behavior patterns or promotional impacts.
5. The North East region is falling behind in both sales and profit, signaling a good opportunity for targeted campaigns or operational improvements.

2. Returns Impact Dashboard

Problem: Returns are negatively impacting profits. Visualize and analyze return rates and reasons.

Tasks:

- Join Returns data with Orders in Tableau.
- Show return rate by category, region, and discount level.
- Create visualizations to identify trends and spikes in returns.
- Suggest actionable insights for reducing returns.



INSIGHTS: -

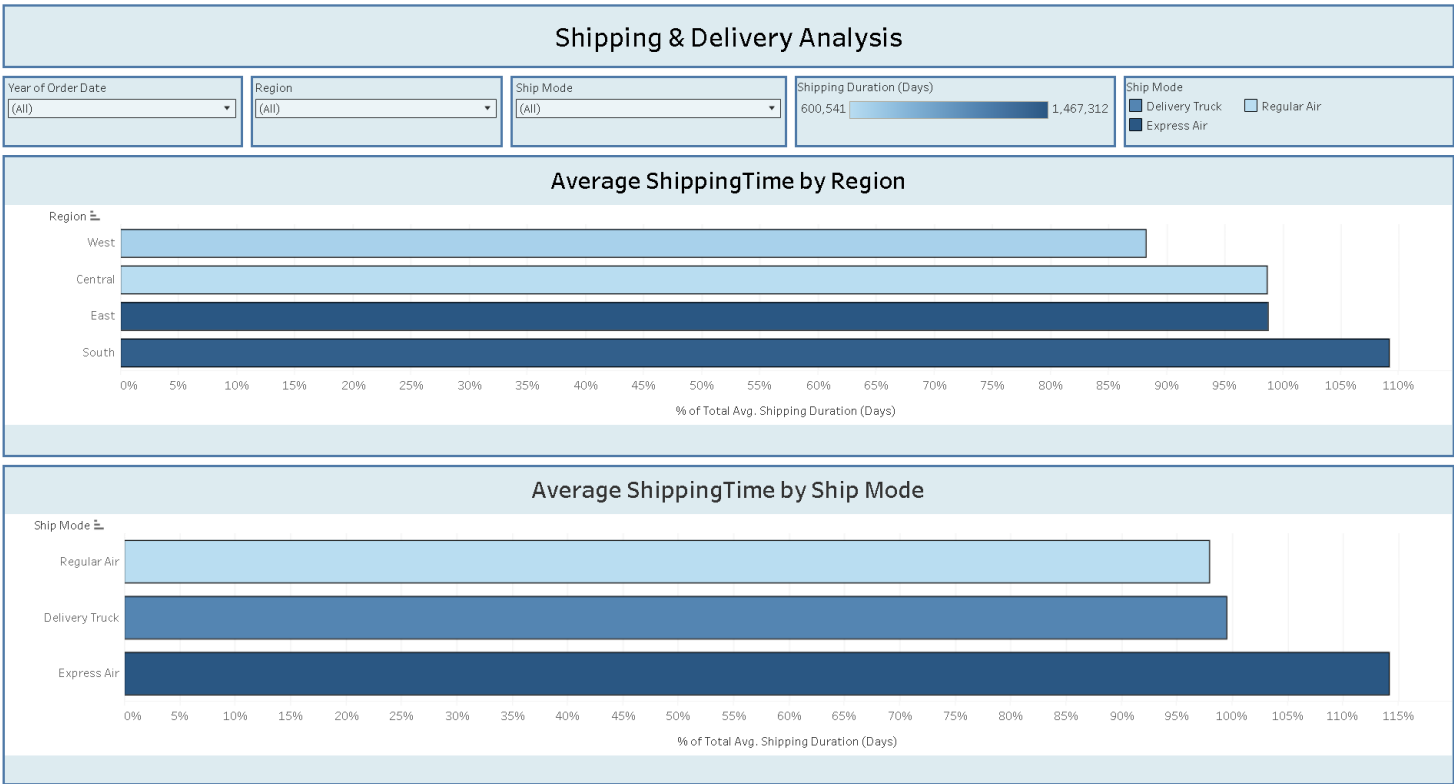
1. Most returns are happening in orders with low discounts, especially in Office Supplies.
2. Central and East regions show a noticeable percentage of returned orders, more than others.
3. Across all discount ranges, Furniture items have very low return rates.
4. There was a visible increase in returns around September–November 2013, hinting at a seasonal or quality issue.
5. The vast majority of orders are successfully delivered without returns, showing overall customer satisfaction is good.

3. Shipping and Delivery Analysis

Problem: Delivery delays may affect customer satisfaction. Analyze shipping times and modes.

Tasks:

- Calculate shipping duration (Ship Date - Order Date) within Tableau.
- Visualize average shipping time by region and shipping method.
- Identify bottlenecks and suggest improvements.
- Build interactive dashboards to explore shipping data.

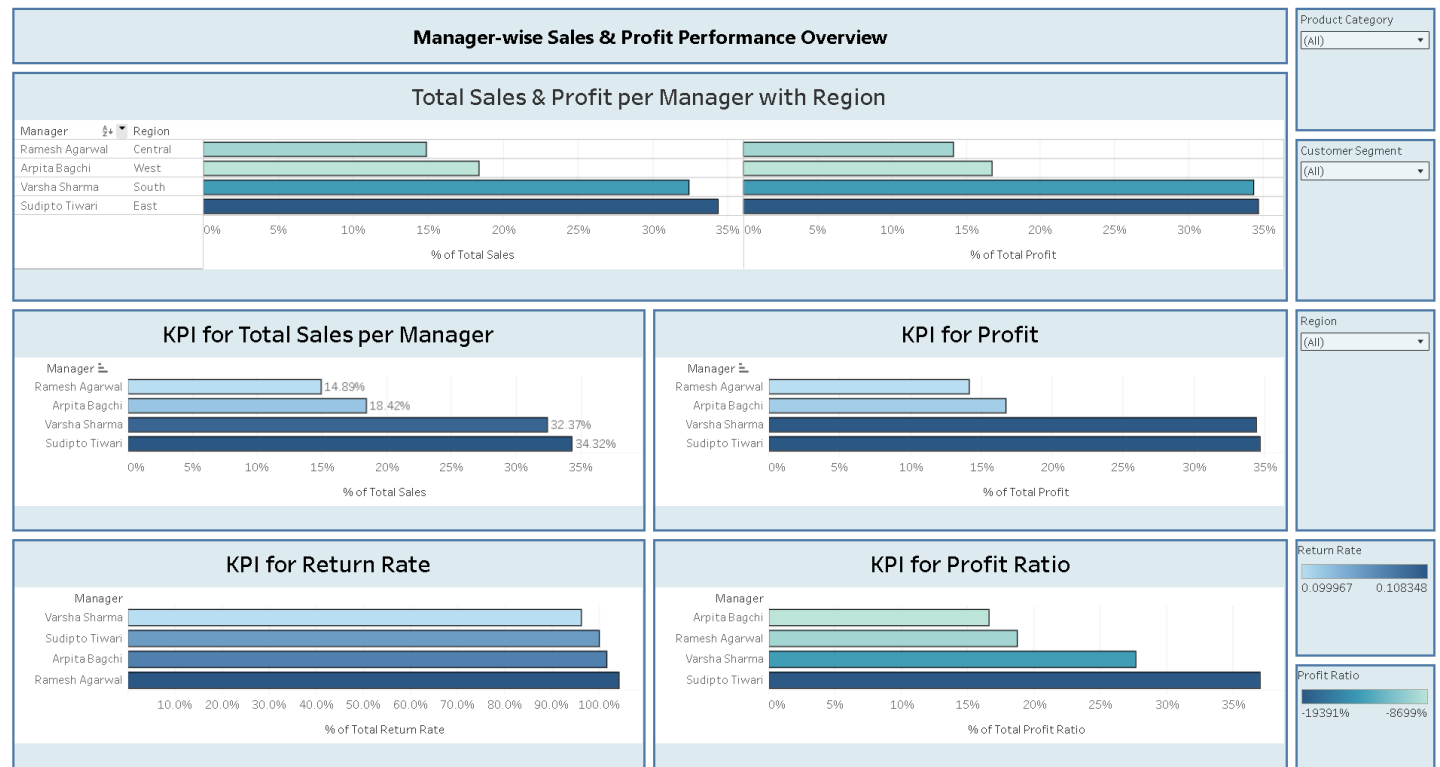


4. Regional Team Performance Analysis

Problem: Evaluate the performance of regional sales managers based on sales and profit metrics.

Tasks:

- Map 'People' data to regions.
- Aggregate sales and profit by region and manager.
- Visualize KPIs such as total sales, profit, return rates by manager.
- Design an easy-to-use manager performance dashboard.



INSIGHTS: -

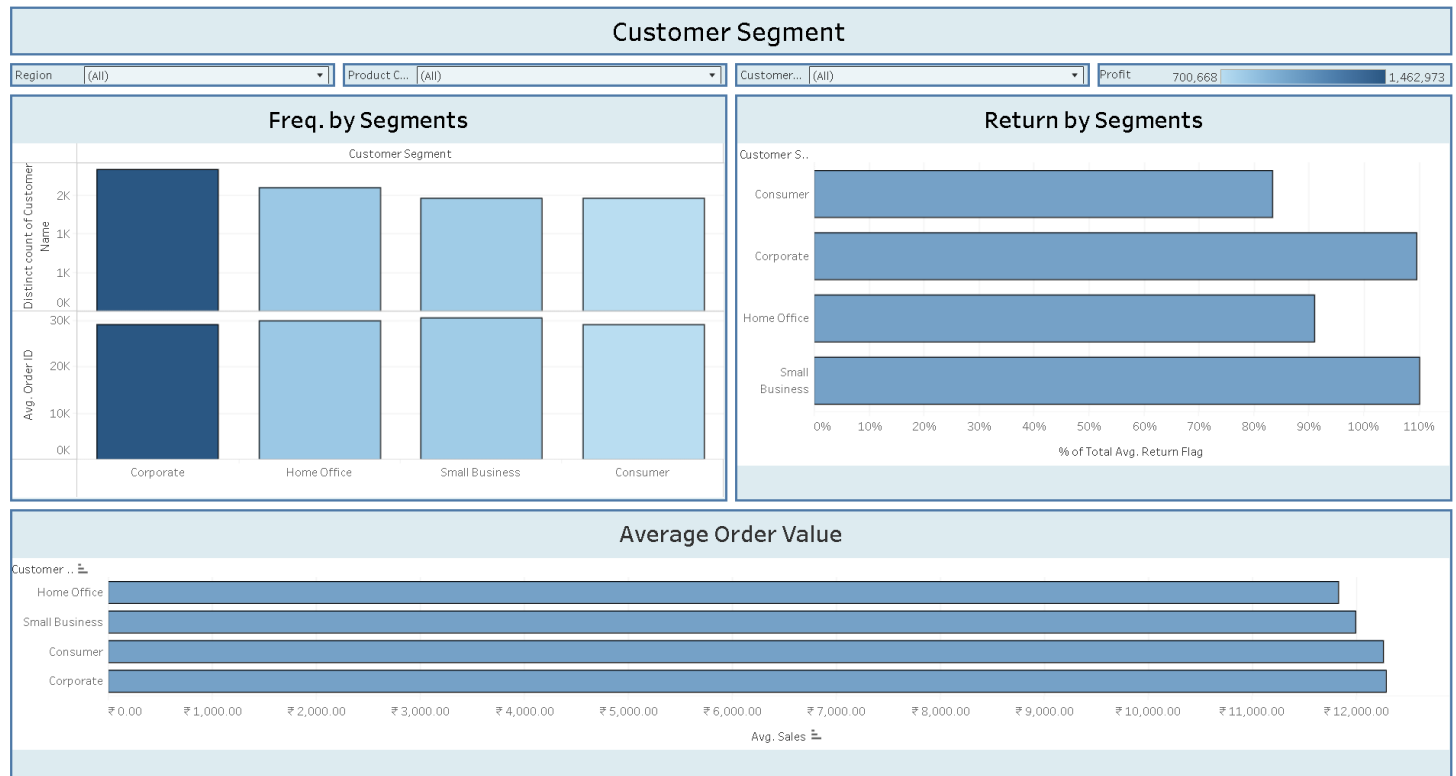
1. Varsha Sharma leads in both total sales and profit, making her the most successful manager overall.
2. Sudipto Tiwari shows strong performance the second-best in both sales and profit, contributing significantly to the company's revenue.
3. Arpita Bagchi Even with lower sales, she has a good profit ratio, showing efficient handling of resources.
4. Ramesh has the lowest sales and profit, and also lags in profit ratio, suggesting a need for support or strategy change.
5. Return rates of managers show a very high return rate, near 100%, which may indicate data issues or product-level concerns.

5. Customer Segment Insights

Problem: Different customer segments show varied buying behavior. Understand these differences.

Tasks:

- Segment customers into Consumer, Corporate, and Home Office.
- Analyze purchase frequency, average order value, and returns by segment.
- Create dashboards showing segment-wise trends and opportunities.
- Use parameters and filters to drill down by segment.



INSIGHTS: -

1. Small Business customers order have the highest purchase frequency, placing the most orders on average.
2. All segments buy almost equally Purchase activity is fairly spread across Consumer, Corporate, Home Office, and Small Business segments.
3. Consumer customers return products the most, crossing 100% return rate, which needs attention.
4. Home Office and Corporate segments show a higher average order value, with Home Office customers spending close to ₹13,000 per order.
5. Small Business makes frequent purchases, their average order value is lower than other segments.

6. Seasonality and Trend Analysis

Problem: Identify seasonal sales trends to support inventory and marketing planning.

Tasks:

- Visualize monthly and quarterly sales trends.
- Identify peak and off-peak periods.
- Create interactive time series visualizations.
- Use forecasting features in Tableau if possible.

